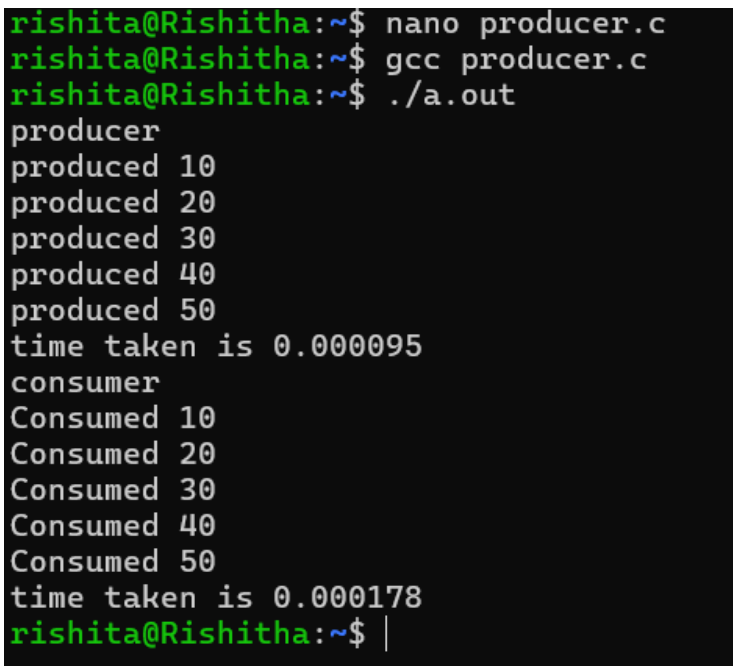


OS-Practical-5

- A producer-consumer program was created using an anonymous pipe.
- The parent process acted as the producer and generated the values 10, 20, 30, 40 and 50.
- The child process acted as the consumer and successfully received the same values through the pipe.
- The program measured the execution time using clock().
- The output showed that the data produced by the parent was successfully consumed by the child process.
- The producer time displayed in the executed output was 0.000095 seconds, while the consumer time was 0.000178 seconds.

Screenshot of the Output



```
rishita@Rishitha:~$ nano producer.c
rishita@Rishitha:~$ gcc producer.c
rishita@Rishitha:~$ ./a.out
producer
produced 10
produced 20
produced 30
produced 40
produced 50
time taken is 0.000095
consumer
Consumed 10
Consumed 20
Consumed 30
Consumed 40
Consumed 50
time taken is 0.000178
rishita@Rishitha:~$ |
```

Shell Command Execution through exec

- Implemented a C program to execute the equivalent of the shell command: `ls -l | grep ".c"`
- The program creates a pipe and uses `fork()` to create a child process.
- The `dup2()` redirects the required standard input/output streams, and `exec()` runs the commands so that the output of `ls -l` is passed to `grep ".c"`.
- The attached terminal screenshot shows the resulting list of C source files.

```

rishita@Rishitha:~$ nano pipeexec.c
rishita@Rishitha:~$ gcc pipeexec.c
rishita@Rishitha:~$ ./a.out
-rw-r--r-- 1 rishita rishita      9 Aug  4 03:00 c2
-rw-r--r-- 1 rishita rishita      0 Aug 24 05:52 copy.txt
-rw-r--r-- 1 rishita rishita     73 Aug 11 05:16 exec1.c
-rw-r--r-- 1 rishita rishita    111 Aug 11 04:59 exec2.c
-rw-r--r-- 1 rishita rishita     89 Aug 11 05:19 exec3.c
-rw-r--r-- 1 rishita rishita     88 Aug 11 05:20 exec4.c
-rw-r--r-- 1 rishita rishita    192 Aug 11 05:34 exec5.c
-rw-r--r-- 1 rishita rishita    392 Sep  1 05:41 fcfs.c
-rw-r--r-- 1 rishita rishita    238 Aug  4 05:07 files.c
-rw----- 1 rishita rishita     54 Aug  3 08:12 files.c.save
-rw-r--r-- 1 rishita rishita    271 Aug  4 04:36 files1.c
-rw-r--r-- 1 rishita rishita    153 Aug  3 09:27 files1.c.save
-rw-r--r-- 1 rishita rishita    182 Sep  6 11:37 fork.c
-rw-r--r-- 1 rishita rishita     81 Aug  1 16:46 fork.c.save
-rw-r--r-- 1 rishita rishita      0 Sep  6 11:37 fork2.c
-rw-r--r-- 1 rishita rishita   387 Aug 24 16:26 fork3.c
-rw-r--r-- 1 rishita rishita   424 Aug 24 16:25 forkprac3.c
-rw-r--r-- 1 rishita rishita     76 Aug 27 04:40 infiniteloop.c
-rw-r--r-- 1 rishita rishita    236 Sep 17 05:18 malloc.c
-rw----- 1 rishita rishita      1 Aug 27 06:34 mutiprocess.c.save
-rw-r--r-- 1 rishita rishita    300 Aug 27 05:42 namedpipeex.c
-rw-r--r-- 1 rishita rishita    350 Aug  1 10:21 open2.c
-rw----- 1 rishita rishita     58 Jul 30 05:38 open2.c.save
-rw-r--r-- 1 rishita rishita    291 Aug 24 05:56 os.c
-rw-r--r-- 1 rishita rishita    251 Aug  7 03:32 pipe.c
-rw-r--r-- 1 rishita rishita    252 Aug  7 03:30 pipe.c.save
-rw-r--r-- 1 rishita rishita    228 Aug 11 05:53 pipeex.c
-rw-r--r-- 1 rishita rishita    246 Sep 24 04:17 pipeexec.c
-rw-r--r-- 1 rishita rishita    251 Aug 11 08:36 pipetask1.c
-rw-r--r-- 1 rishita rishita    251 Aug 11 08:28 pipetask1.c.save
-rw-r--r-- 1 rishita rishita    237 Aug 11 06:12 pipetask2.c
-rw-r--r-- 1 rishita rishita    313 Aug 24 05:09 practical1.c
-rw-r--r-- 1 rishita rishita    312 Aug 24 14:42 practical1.c.save
-rw-r--r-- 1 rishita rishita    337 Sep  1 05:23 process1.c
-rw-r--r-- 1 rishita rishita    255 Sep  1 05:25 process2.c
-rw-r--r-- 1 rishita rishita    600 Sep 24 03:09 producer.c
-rw-r--r-- 1 rishita rishita     88 Aug  3 06:48 read1.c
-rw-r--r-- 1 rishita rishita    230 Aug  3 08:15 readwrite.c

```